EU Compliance-as-a-Service: Battery Passport & Textile Regulation Products for Indian Manufacturers

Executive Summary

The European Union's regulatory framework is undergoing its most sweeping transformation in decades, creating both existential compliance risk and massive product-market opportunity for Indian manufacturers. Two regulations — the **EU Battery Regulation 2023/1542** and the **Ecodesign for Sustainable Products Regulation (ESPR) 2024/1781** — will fundamentally reshape how Indian batteries and textiles enter the EU market. The Digital Battery Passport becomes mandatory on **18 February 2027**, while the textile Digital Product Passport follows in **late 2028/early 2029**. Combined, Indian exports in these categories to the EU exceed **$8 billion annually** and are projected to reach **$20+ billion by 2030** following the India-EU FTA's zero-duty provisions. (Keypoint Intelligence) The penalty for non-compliance is severe: market access denial at EU borders, fines up to **4% of EU turnover**, public naming in the EU Safety Gate database, and de-listing from online marketplaces. (InfoDPP) Yet the vast majority of Indian manufacturers — particularly the SME segment that constitutes over 80% of India's textile export base — lack the technical infrastructure, regulatory expertise, and capital to build compliance systems in-house. This report presents two distinct product concepts: a **Battery Passport Compliance Platform** and a **Textile EU Compliance Platform**, architected specifically for Indian manufacturers selling into the EU. Each product is designed as a multi-tenant SaaS solution that turns regulatory burden into competitive advantage by automating data collection, generating compliant Digital Product Passports, managing multi-regulatory workflows, and providing audit-ready documentation.

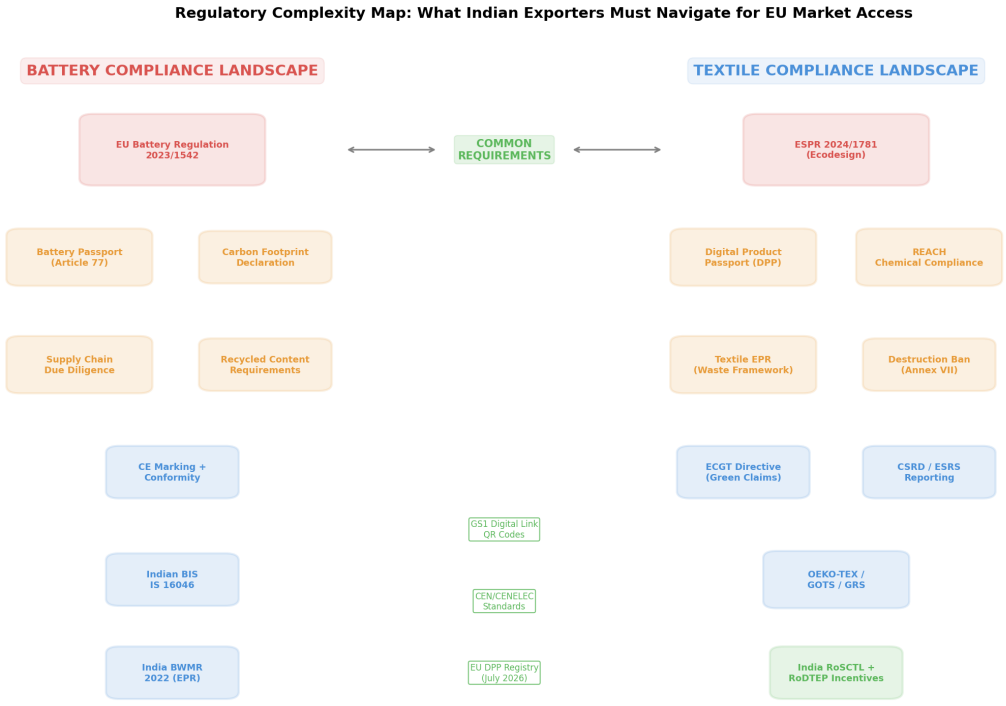


Figure 1: Regulatory Complexity Map

1. Product I: Battery Passport Compliance Platform

1.1 Regulatory Foundation: EU Battery Regulation 2023/1542

The EU Battery Regulation 2023/1542 entered into force in **August 2023** and replaces the earlier Battery Directive 2006/66/EC. Unlike directives, this is a directly applicable regulation across all 27 EU member states, meaning no national transposition is required — the rules are uniform and enforceable immediately. (TraceX) For Indian manufacturers, this means the compliance bar is identical whether exporting to Germany, France, or Poland. The regulation introduces a comprehensive lifecycle approach to battery governance, covering everything from raw material extraction through manufacturing, use, and end-of-life recycling. The regulation applies to all batteries placed on the EU market, but the most stringent requirements — including the Digital Battery Passport — apply specifically to **electric vehicle (EV) batteries**, **light means of transport (LMT) batteries**, and **industrial batteries with capacity above 2 kWh**. (carbmee.com) Portable batteries are subject to lighter requirements but are not exempt from all obligations. The critical implication for Indian exporters is that the passport requirement is tied to placement on the EU market, not to the location of manufacture. A battery produced in Tamil Nadu and sold into the EU through a distributor in Rotterdam must carry a compliant battery passport. (carbmee.com)

Battery Category	Passport Required?	Key Deadlines	Indian Export Relevance
EV Batteries	**Yes** —Mandatory	Passport: Feb 2027; Carbon footprint: Feb 2026	High —EV supply chain expanding rapidly
LMT Batteries (e-bikes, e-scooters)	**Yes** —Mandatory	Passport: Feb 2027	Medium —Growing segment for Indian manufacturers
Industrial Batteries >2 kWh	**Yes** —Mandatory	Passport: Feb 2027; Carbon footprint: phased	High —Energy storage, industrial applications
Portable Batteries	No passport (yet)	Removability: Feb 2027	Medium —Consumer electronics, smartphones
SLI Batteries	No passport	Labelling requirements apply	Low —Mostly automotive OEM supply

The regulation's implementation follows a carefully sequenced timeline designed to give industry preparation time while progressively tightening requirements. **Carbon footprint declarations** became mandatory in February 2026, requiring manufacturers to calculate and declare the product-level carbon footprint of each battery model per manufacturing plant, expressed in kg CO2 equivalent per kWh of total energy delivered over the battery's expected service life. (psqr.eu) The calculation must cover four lifecycle stages: raw material acquisition and pre-processing, manufacturing, distribution, and end-of-life and recycling. Company-specific activity data is mandatory for the manufacturing and distribution stages, while secondary data may be used for other stages. (The Battery Pass) **Labelling requirements** take effect in August 2026, requiring visible labels indicating expected lifetime, capacity, chemical composition, separate collection symbols, and presence of hazardous substances. (psqr.eu) The **Digital Battery Passport** becomes mandatory on **18 February 2027**, and this is the single most complex technical requirement that Indian manufacturers must prepare for. Due diligence obligations, originally scheduled for August 2025, have been postponed to **August 2027** following the Omnibus IV proposal, giving companies additional time to map their cobalt, lithium, and nickel supply chains. (psqr.eu)

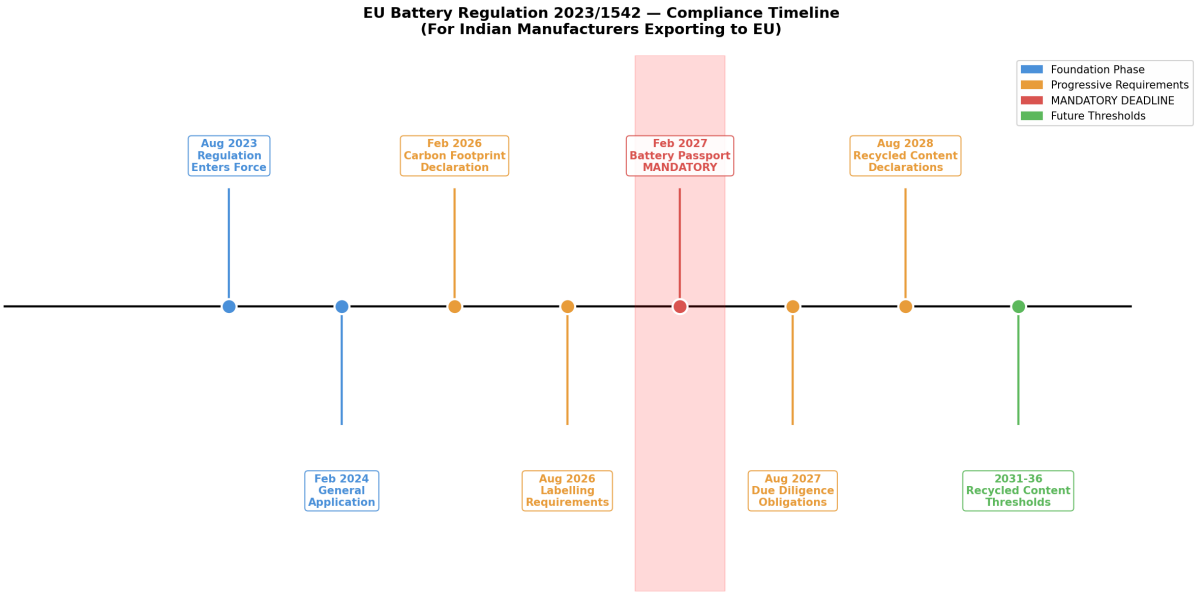


Figure 2: Battery Compliance Timeline

1.2 What the Battery Passport Must Contain

The Digital Battery Passport is not merely a document — it is a structured, machine-readable digital record linked to each individual battery through a unique identifier, typically accessed via **QR code**. The passport's content requirements are defined by **DIN DKE SPEC 99100**, published in January 2025 and updated to version 1.3 in March 2026, which establishes the data attributes to be included in the battery passport based on EU Battery Regulation requirements as well as voluntary additions. (The Battery Pass) The specification aligns with CEN/CENELEC JTC 24 drafts and integrates ongoing implementing acts under the EU Battery Regulation. The data model is formally expressed as an Asset Administration Shell (AAS) submodel, with semantic IDs tied to ECLASS reference data, enabling machine-to-machine interoperability across the value chain. (IDTA) The content is organized into **six mandatory data categories**, each with specific fields, access levels (public vs. restricted), and update frequencies (static vs. dynamic).

Data Category	Key Data Fields	Access Level	Update Type	DIN DKE SPEC Reference
General Battery & Manufacturer Info	Manufacturer name, model ID, batch/ serial number, manufacturing date/ place, battery category, mass	Public	Static	6.1.2, 6.1.3
Material Composition & Sourcing	Battery chemistry, critical raw materials (>0.1% w/ w), hazardous substances, recycled content %, due diligence status	Mixed (public + restricted)	Static + Dynamic	6.5

Table 2 – continued

Data Category	Key Data Fields	Access Level	Update Type	DIN DKE SPEC Reference
Performance & Durability	Rated capacity, nominal voltage, energy throughput, cycle life, capacity fade, round-trip efficiency, state of health	Public (model-level) + PLI (item-level)	Static + Dynamic	6.7
Carbon Footprint	Total kg CO2e/kWh, breakdown by lifecycle stage, performance class, declaration URL	Public	Static	6.3
Safety Information	Extinguishing agents, safe handling instructions, transport classification, hazard symbols	Public	Static	6.2
End-of-Life & Recycling	Disassembly instructions, material recovery targets, recycling guidance, waste classification	Public + Restricted	Dynamic	6.6

The **public data layer** is accessible to anyone scanning the QR code — consumers, recyclers, customs officials, and competitors. This includes basic identification, performance specifications, carbon footprint summary, and safety information. The **restricted data layer** is accessible only to regulators, authorized representatives, and persons with legitimate interest (PLI) — this includes commercially sensitive supply chain details, detailed due diligence documentation, and dynamic operational data like state of health. The **dynamic data layer** updates over the battery's lifecycle — state of health, cycle count, remaining capacity, and service history change as the battery is used, maintained, repaired, or repurposed. (carbmee.com) This three-tier access model is central to the product architecture because it requires granular permission management, audit logging, and role-based access control (RBAC) that the platform must implement.

1.3 Indian Regulatory Context: Dual Compliance Burden

Indian battery manufacturers face a **dual compliance burden**: satisfying EU Battery Regulation requirements for export markets while simultaneously complying with India's domestic regulatory framework. Understanding this dual burden is critical to product design because the platform must serve both compliance streams, not just the EU side. India's **Battery Waste Management Rules 2022 (BWMR)**, administered by the Central Pollution Control Board (CPCB), establish Extended Producer

Responsibility (EPR) for all battery categories — EV, portable, automotive, and industrial. (IEA – International Energy Agency) Every producer, importer, and recycler must register on the CPCB EPR portal and meet collection and recycling targets that escalate annually. For EV batteries specifically, producers must achieve **70% collection** of batteries placed on the market by 2027-28, with recovery targets increasing from **70% in 2024-25 to 90% by 2026-27**. (IEA – International Energy Agency) From FY 2027-28, new batteries manufactured in India must incorporate minimum percentages of domestically recycled materials — starting at **5% and scaling to 20% by 2030-31**. (Gravita India Ltd) The 2025 amendments to BWMR formally recognize second-life battery applications (such as stationary energy storage) as valid pre-recycling pathways, and introduce recycled content requirements that align conceptually with the EU's recycled content mandates, though the specific thresholds and timelines differ. (Gravita India Ltd)

On the safety certification front, the **Bureau of Indian Standards (BIS)** mandates **IS 16046:2018** certification for sealed secondary lithium cells and batteries, harmonized with IEC 62133-2:2017. (absoluteveritas.com) A February 2026 update introduced mandatory **rated capacity verification** testing per IS 16047 (Part 3):2018, effective May 2027 for all new registrations, inclusions, and renewals. (gmalabs.com) This creates a direct data point — verified rated capacity — that flows into the EU Battery Passport's performance data section. The platform must therefore integrate BIS certification data as an input stream. Additionally, India's **PLI Scheme for Advanced Chemistry Cell (ACC) Battery Storage**, approved in May 2021 with an outlay of 18,100 crore ($2.08 billion), targets **50 GWh of domestic ACC manufacturing capacity**. (Press Information Bureau) The scheme incentivizes domestic value addition (25% within 2 years, 60% within 5 years) and directly impacts the supply chain traceability data that EU due diligence requirements demand. However, as of October 2025, only **1.4 GWh (2.8%) of targeted capacity** had been commissioned, and **zero incentives** had been disbursed due to beneficiaries failing to meet commercial production and DVA thresholds. (Down To Earth) This reality — that Indian cell manufacturing remains almost entirely import-dependent, primarily from China — creates acute due diligence challenges for Indian battery exporters who must now trace and document the origin of cells and raw materials.

Indian Regulation	Key Requirements	EU Alignment	Platform Integration Point
Battery Waste Management Rules 2022	EPR registration, collection/recycling targets, recycled content mandates	Recycled content % maps to EU Passport	CPCB EPR portal API integration
BIS IS 16046:2018 + IS 16047	Safety certification, rated capacity verification	Verified capacity feeds EU performance data	BIS certification database connector
PLI ACC Scheme	50 GWh target, 60% DVA requirement, subsidy per kWh	DVA tracking supports EU due diligence	PLI compliance dashboard module
MoEFCC Environmental Clearance	EIA, pollution control, hazardous waste	Environmental data for EU carbon footprint	Environmental data ingestion pipeline

1.4 Product Architecture: Battery Passport Compliance Platform

The Battery Passport Compliance Platform is designed as a **four-layer SaaS architecture** that transforms scattered, siloed compliance data into a unified, audit-ready Digital Battery Passport. The architecture is specifically optimized for the Indian manufacturing context — ERP systems like Tally, SAP

Business One, and custom-built solutions; supplier ecosystems dominated by SMEs with limited digital maturity; and regulatory reporting requirements spanning both Indian (CPCB) and EU authorities.

1.4.1 Layer 1: Data Collection & Integration The data collection layer is the foundation of the entire platform. Its purpose is to ingest data from every system and stakeholder that contributes to the battery passport, normalize it into the DIN DKE SPEC 99100 data model, and maintain data provenance for audit purposes. The layer exposes a set of **pre-built connectors** for common Indian manufacturing IT systems: SAP ERP, Oracle NetSuite, Tally Prime, Microsoft Dynamics, and custom SQL/NoSQL databases. For PLM systems, connectors support Siemens Teamcenter, Dassault ENOVIA, and PTC Windchill. The connector framework is extensible through a **plugin SDK** that allows integration partners to build custom connectors for proprietary systems common among Indian SMEs.

Supplier data collection is the most challenging aspect of battery passport implementation because much of the required information — raw material origins, carbon intensity of upstream processes, due diligence documentation — resides with **tier-2 and tier-3 suppliers** who may not have digital systems at all. The platform addresses this through a **Supplier Portal** — a lightweight, mobile-responsive web application that suppliers access via invitation. The portal presents structured data entry forms mapped to DIN DKE SPEC 99100 fields, supports document upload (PDF certificates, test reports, SDS sheets), and provides real-time validation feedback. For suppliers without reliable internet, the platform supports **offline data capture** through Excel templates that can be uploaded in bulk, with automated validation and error flagging. The supplier onboarding workflow includes **multi-language support** (Hindi, Tamil, Telugu, Gujarati, Kannada, and English) to accommodate India's linguistic diversity.

IoT integration captures dynamic battery data during manufacturing and testing. The platform integrates with battery test equipment (Arbin, MACCOR, Bitrode) via OPC-UA and MQTT protocols to automatically capture capacity, internal resistance, cycle life, and degradation data directly from test stations. This eliminates manual data entry errors and ensures test data flows directly into the passport's performance section with tamper-evident timestamps. For operational batteries in the field, the platform can ingest BMS (Battery Management System) telemetry — state of charge, state of health, temperature profiles, cycle counts — via secure APIs, enabling the dynamic data layer of the passport to update in real time.

Data Source	Integration Method	Data Captured	Frequency
ERP (SAP/Tally/ Dynamics)	REST API / ODBC	Production volumes, batch records, material consumption	Real-time / Daily sync
PLM (Teamcenter/ ENOVIA)	REST API	BOM, technical specifications, design revisions	Event-driven
Supplier Portal	Web app + Excel bulk upload	Raw material origins, certificates, due diligence docs	On submission + periodic refresh
Test Equipment (Arbin/ MACCOR)	OPC-UA / MQTT	Capacity, resistance, cycle life, degradation	Per test cycle
BMS / IoT	Secure API / MQTT	SoC, SoH, temperature, cycle count, fault codes	Continuous (optional)
BIS Certification DB	API / Manual upload	IS 16046 certificate, rated capacity verification	Per certification cycle

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Table 4 – continued

Data Source	Integration Method	Data Captured	Frequency
CPCB EPR Portal	Web scraping / API	EPR registration, collection targets, recycler credits	Monthly sync

1.4.2 Layer 2: Compliance Engine The compliance engine is the brain of the platform — it takes raw data from Layer 1 and transforms it into regulatory-compliant outputs through a series of specialized modules. Each module is designed around a specific regulatory requirement and implements the relevant calculation methodology, validation rules, and documentation format.

The **DIN DKE SPEC 99100 Mapper** is the core module that validates all ingested data against the specification's data attribute requirements. It performs semantic validation (data types, units, ranges), referential integrity checks (cross-field consistency), and completeness scoring (what percentage of mandatory fields are populated). The mapper generates a **Compliance Score** for each battery model — a weighted metric indicating passport readiness. A score below 100% triggers automated alerts to the responsible compliance team member with specific gap identification.

The **Carbon Footprint Calculator** implements the EU Commission's delegated act methodology for battery carbon footprint calculation. The module collects activity data (energy consumption, material inputs, transport distances) and applies emission factors from the GBA GHG Rulebook and the Battery Pass Carbon Footprint Rules to compute cradle-to-grave carbon intensity in kg CO2e/kWh. (The Battery Pass) It supports both primary data (company-specific measurements) and secondary data (industry averages) with clear documentation of data quality tiers, as required by the EU methodology. The calculator produces the structured carbon footprint declaration format that must accompany the battery until February 2027 and thereafter be accessible via the passport QR code. (psqr.eu) Critically, the module **explicitly excludes carbon offsets** from the declared footprint, as the EU regulation mandates, while allowing offsets to be reported separately as supplementary information.

The **Due Diligence Mapper** supports compliance with Article 47-50 of the Battery Regulation, which requires supply chain due diligence policies aligned with international responsible sourcing standards. The module implements the **OECD Due Diligence Guidance for Responsible Supply Chains** framework, guiding users through risk identification (country of origin, red-flag indicators), risk assessment (severity and likelihood scoring), risk mitigation (action plans and monitoring), and third-party verification. The module maintains a **Supplier Risk Register** that tracks each supplier's due diligence status, flags high-risk origins (conflict-affected areas, areas with severe human rights abuses), and generates the documentation required for audit. Given that Indian battery manufacturers source cells and raw materials predominantly from China — which carries elevated due diligence risk under EU criteria — this module is particularly critical.

The **Recycled Content Tracker** monitors the percentage of recycled lithium, cobalt, nickel, and lead in each battery model. It integrates with recycler certification data, tracks material flows from waste batteries through reprocessing to new cell production, and calculates recycled content percentages according to the EU methodology (expected in implementing acts during 2026). The tracker connects to India's CPCB EPR portal to incorporate domestic recycling credits and ensure alignment with Indian BWMR recycled content mandates.

1.4.3 Layer 3: Passport Generation & QR Code Management The passport generation layer creates the actual Digital Battery Passport — the structured, machine-readable output that regulators, customers, and recyclers will access. The platform generates passports in **multiple formats** to serve different use cases: the AAS (Asset Administration Shell) JSON format for machine-to-machine data exchange, a human-readable HTML landing page for QR code resolution, and PDF certificates for offline distribution.

The **QR Code Generator** implements the **GS1 Digital Link** standard, which is the recommended data carrier technology for Digital Product Passports under ESPR. (Linkode) GS1 Digital Link encodes the product's GTIN (Global Trade Item Number) as a web URI within a QR code. When scanned by a consumer smartphone, the QR code resolves to the passport's public-facing HTML page. When scanned by a retail POS system (post-Sunrise 2027), the same QR code extracts the GTIN for checkout processing. (QRStuff) When accessed by a regulator or recycler through authenticated channels, the resolver returns the appropriate data tier (public, restricted, or dynamic) based on the requester's identity and permissions. The platform provides **dynamic QR codes** — the physical code printed on the battery never changes, but the data it resolves to can be updated as regulations evolve or as dynamic data changes, eliminating repackaging costs.

The **AAS Submodel Builder** constructs the battery passport as an IEC 63278-compliant Asset Administration Shell submodel. This format is essential for industrial interoperability — it allows the passport to be consumed by OEM systems, recycling facility software, and regulatory databases in a standardized, machine-interpretable structure. The builder maps all DIN DKE SPEC 99100 data attributes to their corresponding semantic IDs in the ECLASS taxonomy, ensuring that any system reading the passport can unambiguously interpret each data field. (IDTA)

Passport Output Format	Use Case	Target Audience	Technical Standard
AAS JSON Submodel	Machine-to-machine exchange, registry submission	Regulators, OEMs, recyclers	IEC 63278, DIN DKE SPEC 99100
GS1 Digital Link QR Code	Physical product label, consumer access	Consumers, customs, retailers	GS1 Digital Link URI syntax
HTML Landing Page	Human-readable passport display	Consumers, auditors, buyers	Responsive web, multi-language
PDF Certificate	Offline distribution, audit documentation	Compliance officers, customs	PDF/A, digitally signed
API Response (JSON/ XML)	System integration, batch queries	ERP, PLM, marketplace platforms	REST API, OAuth 2.0

1.4.4 Layer 4: Registry Integration & Reporting The registry integration layer ensures that generated passports are properly registered, reported, and maintained across all relevant regulatory systems. The **EU DPP Registry API Connector** submits product identifiers, passport URLs, and conformity declarations to the EU Central DPP Registry, which goes live in **July 2026**. (openProd.io) The connector handles authentication, data validation against registry schema, and submission confirmation tracking. It also monitors for registry updates — when the EU publishes new implementing acts or modifies data requirements, the connector flags affected passports and triggers update workflows.

The **CPCB EPR Portal Connector** automates Indian compliance reporting by extracting relevant data from the platform and formatting it for the Central Pollution Control Board's EPR portal. This includes

battery placement volumes, collection target calculations, recycler credit purchases, and environmental compensation payments. The connector supports both API integration (where available) and structured Excel export for manual upload.

The **Audit & Documentation module** generates comprehensive compliance reports for internal audit, third-party verification, and regulatory inspection. Reports include data provenance logs (who entered what data, when, from what source), compliance score trends over time, gap analyses with remediation recommendations, and evidence packages for each passport data point. All reports are digitally signed and tamper-evident, using blockchain anchoring for critical audit trails.

1.5 Feature Roadmap & Go-to-Market

Phase	Timeline	Features	Target Segment
MVP	Q3 2026	Supplier portal, basic passport generation, QR codes, carbon footprint calculator	Large battery OEMs (Ola Electric, Exide, Amara Raja)
Phase 2	Q1 2027	Full DIN DKE SPEC 99100 compliance, AAS submodel export, EU DPP Registry integration, due diligence mapper	Mid-size manufacturers, EV battery suppliers
Phase 3	Q3 2027	IoT/BMS integration, dynamic data updates, predictive compliance analytics, marketplace integrations	Full market, SME segment via self-service
Phase 4	2028+	Multi-product DPP (beyond batteries), AI-powered supplier risk scoring, carbon optimization recommendations	Platform expansion to adjacent product categories

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2. Product II: Textile EU Compliance Platform

2.1 Regulatory Foundation: ESPR and the Textile Regulatory Stack

The Ecodesign for Sustainable Products Regulation (ESPR) 2024/1781 entered into force on **18 July 2024** and represents the EU's most ambitious product sustainability framework to date. (complir.io) Unlike the previous Ecodesign Directive (2009/125/EC), which covered only energy-related products, ESPR applies to **virtually all physical products** placed on the EU market — including textiles, furniture, steel, aluminium, tyres, and mattresses. The regulation operates as a framework: it establishes the legal basis and rules, while specific product requirements are defined through **delegated acts** adopted product group by product group. Textiles and apparel are designated as a **priority product group** in the ESPR Working Plan 2025-2030, with the textile delegated act expected for adoption in **Q2 2027**

and compliance becoming mandatory approximately **18 months later — late 2028 or early 2029**. (asuene.com)

However, the textile delegated act is only one layer of a much broader regulatory stack that Indian textile exporters must navigate. The full stack includes **REACH** (chemical substance restrictions), the **Waste Framework Directive** (textile EPR requirements), the **ECGT Directive** (green marketing claims), **CSRD/ESRS** (sustainability reporting), and the **EU-India FTA** (tariff elimination with compliance conditionality). Understanding the interactions between these regulations is essential because the Textile EU Compliance Platform must address them as an integrated system, not as isolated requirements.

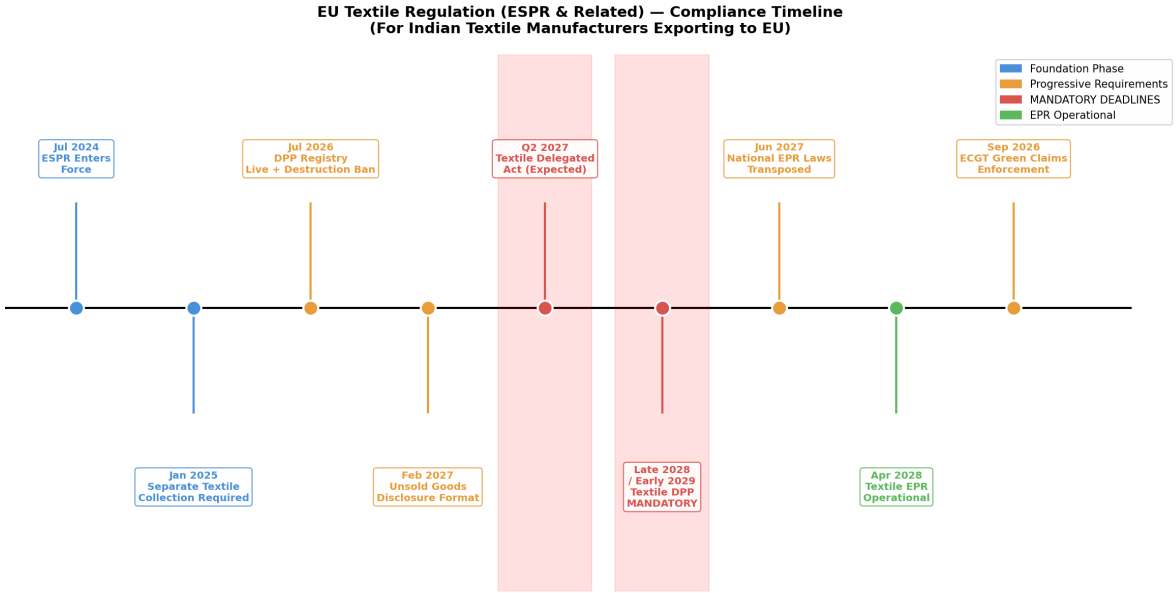


Figure 3: Textile Compliance Timeline

Regulation	Scope	Status	Enforcement Date	Indian Exporter Impact
ESPR 2024/1781	Ecodesign, DPP, destruction ban	In force	DPP: ~Late 2028/2029	**Critical** – product-level data passport mandatory
REACH (EC 1907/2006)	Chemical restrictions (SVHCs, azo dyes, phthalates)	In force	Ongoing	**Critical** – non-compliance = market denial
Waste Framework Directive + Textile EPR	Extended producer responsibility, collection, recycling	Adopted Oct 2025	EPR operational: April 2028	**High** –fees based on volume and eco-modulation
ECGT Directive 2024/825	Bans generic green claims, offset-based neutrality claims	Adopted	Enforcement: Sep 2026	**High** –marketing compliance required
CSRD / ESRS	Sustainability reporting for large companies	Delayed (Omnibus I)	Non-EU: FY2028 (report 2029)	**Medium** –applies to large Indian groups with EU turnover >€450M

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Table 7 – continued

Regulation	Scope	Status	Enforcement Date	Indian Exporter Impact
EU-India FTA	Zero-duty market access	Signed Jul 2025	In force early 2026	**Opportunity** – compliance enables competitive advantage

2.2 The Digital Product Passport for Textiles

The Digital Product Passport (DPP) is the central information infrastructure of ESPR. While the exact data fields for textiles will only be confirmed when the delegated act is published, the JRC (Joint Research Centre) preparatory study published its **3rd milestone in December 2025**, providing the most concrete picture yet of expected requirements. (carbonfact.com) The study covers apparel containing at least **80% textile fibers**, including workwear and sportswear. Footwear is currently out of scope for the first delegated act, with a separate study expected by end of 2027.

The JRC's lifecycle assessment found that **raw material production accounts for 60-63% of a garment's environmental impact**, manufacturing for 21-29%, and use stage/end-of-life for minor shares. (carbonfact.com) Across all 16 Product Environmental Footprint (PEF) impact categories, the three largest contributors are **water use, climate change, and fossil resource use**. This finding directly shapes the expected DPP data requirements — material composition and origin will be the most heavily weighted data fields.

Expected DPP Data Category	Specific Data Points	Data Source	Confidence Level
Product Identification	GTIN, product name, brand, model, size, color	Manufacturer ERP/PLM	High –standard across DPPs
Material Composition	Fiber content (% by weight), material origins, recycled content %	Supplier declarations, testing labs	High –JRC LCA shows 60%+ impact
Manufacturing Footprint	PEFCR manufacturing stage score, benchmark comparison	Primary data from facilities	Medium –proposed as voluntary excellence indicator
Circularity Attributes	Durability index, repairability score, recyclability %, recycled content	Design data, testing, certification	High –core ESPR objective
Chemical Compliance	REACH SVHC screening, restricted substance testing (azo dyes, phthalates, PFAS)	Laboratory testing (OEKO-TEX, etc.)	High –existing requirement
Supply Chain Transparency	Tier 1-3 supplier identities, facility locations, audit status	Supplier onboarding, audit platforms	Medium-High –aligned with CSDDD

Table 8 – continued

Expected DPP Data Category	Specific Data Points	Data Source	Confidence Level
EPR & End-of-Life	Eco-modulation fee category, take-back instructions, recycling guidance	National EPR scheme rules	Medium —depends on national implementation

The JRC has also proposed **potential minimum recycled-content thresholds** as part of future performance requirements, though these remain under assessment: up to **20% recycled cotton in denim**, up to **15% recycled polyester in knitted and woven products**, up to **10% recycled wool**, and up to **5% recycled nylon**. (carbonfact.com) These thresholds, if adopted in the delegated act, would become binding compliance requirements, not merely disclosure items. The manufacturing footprint indicator is proposed as a voluntary disclosure using the PEFCR methodology limited to the manufacturing stage — brands that beat the category benchmark can display this on the DPP as an excellence indicator, creating a competitive differentiation mechanism.

2.3 REACH and Chemical Compliance: The Entry Gate

While the DPP is the headline requirement, **REACH compliance remains the entry gate** for Indian textiles into the EU market. REACH (Registration, Evaluation, Authorisation and Restriction of Chemicals) has been in force since 2007 and controls the use of chemical substances in products imported into the EU. For textiles, REACH Annex XVII restricts specific substances, while the Candidate List of SVHCs (Substances of Very High Concern) requires disclosure when SVHCs exceed 0.1% w/w in any article component. (Centexbel)

The textile-specific restricted substances under REACH include **azo dyes** that can release carcinogenic aromatic amines (restricted to 30 mg/kg), **disperse dyes** known to cause allergic contact dermatitis, **phthalates** (DEHP, BBP, DBP, DIBP restricted to 0.1% sum total), **PFAS** (per- and polyfluoroalkyl substances, increasingly restricted), **heavy metals** (cadmium, lead, chromium, nickel with specific migration limits), and **flame retardants** (PBBs, PentaBDE, OctaBDE prohibited). (europa.eu) The EU Ecolabel's Restricted Substance List provides the most comprehensive reference for textile chemical compliance, covering additional categories including biocides, alkylphenol ethoxylates (APEOs), halogenated carriers, and chlorophenols. (europa.eu)

For Indian exporters, REACH compliance is not optional — customs authorities at EU ports actively check for chemical compliance, and non-compliant shipments are stopped, held, refused entry, or destroyed at the importer's expense. (InfoDPP) The **EU Safety Gate** database publicly lists non-compliant products and companies, creating reputational damage that extends beyond the immediate shipment loss. Enforcement is particularly aggressive in **Germany**, where market surveillance authorities actively inspect textile imports and have the power to issue immediate sales bans. (Alibaba.com Seller Central)

Substance Category	Restricted Substances	Limit Value	Testing Method	Risk for Indian Manufacturers
Azo Dyes (releasing aromatic amines)	22 specific amines (4-aminodiphenyl, benzidine, etc.)	30 mg/kg each	EN ISO 14362-1 / DIN 54231	High —azo dyes still widely used in India

Table 9 – continued

Substance Category	Restricted Substances	Limit Value	Testing Method	Risk for Indian Manufacturers
Disperse Dyes	Disperse Blue 1, 3, 7, 26, 35, 102, 106, 124; Disperse Orange 1, 3, 37/76; Disperse Red 1, 11, 17; Disperse Yellow 1, 3, 9, 39, 49	Various	DIN 54231 / HPLC-DAD	Medium — sensitizing dyes in synthetic fiber processing
Phthalates	DEHP, BBP, DBP, DIBP	0.1% sum total	GC-MS	Medium —plastic accessories, prints
PFAS	PFOA, PFOS, PFHxA (expanding list)	1-25 g/m² or ppb	LC-MS/MS	High — water-repellent treatments
Heavy Metals	Cadmium, lead, chromium VI, nickel, arsenic	0.1-60 mg/kg (varies)	ICP-MS, AAS	Medium —metal accessories, dyes
Flame Retardants	PBBs, PentaBDE, OctaBDE, HBCDD, SCCP	Prohibited	GC-MS	Low —limited use in apparel

2.4 The Destruction Ban and EPR: Operational Game-Changers

Two requirements fundamentally alter the operational model of textile exporters: the **destruction ban on unsold goods** and **Extended Producer Responsibility (EPR)** for textile waste. The ESPR's destruction ban, applicable from **19 July 2026 for large companies** (250+ employees, €50M+ turnover), prohibits the destruction of unsold apparel, clothing accessories, and footwear. (Productwise) "Destruction" under EU law includes recycling, energy recovery, and disposal — so companies cannot simply redirect unsold inventory to recycling. Instead, they must prioritize alternatives: resale, remanufacturing, donation, or reuse. The ban carries **10 specific derogations** where destruction remains permitted — dangerous products, legally non-compliant goods, IP-infringing items, technically irreparable products, and goods that cannot be donated after reasonable attempts (documented by offering to at least three social economy entities or listing on an accessible website for at least 8 weeks without acceptance). (anthesisgroup.com)

For Indian exporters, the destruction ban creates indirect pressure even if they are not themselves large companies placing products on the EU market. EU brand customers — H&M, Zara, Primark, Decathlon — are directly subject to the ban and will increasingly require their Indian suppliers to provide **accurate demand forecasting, smaller minimum order quantities, and flexible production scheduling** to minimize overproduction. The platform must therefore include a **Demand-Production Alignment module** that helps Indian manufacturers optimize order quantities against forecasted demand, reducing the risk of unsold inventory accumulating in EU warehouses.

The **EU Textile EPR Directive**, formally adopted in October 2025, requires all member states to establish textile EPR schemes by **April 2028**. (ecosistant) Under EPR, producers (including non-EU brands selling into the EU) bear the costs of collecting, sorting, and recycling textile waste. Fees are

eco-modulated — adjusted based on the environmental performance of the textile product, including durability, recyclability, recycled content, and fast fashion practices. (Sedgwick) Member states may impose **higher fees on fast fashion** and products with poor circularity performance. The EPR obligations apply to clothing, footwear, accessories, household linens, curtains, and potentially mattresses. Online sellers must register with Producer Responsibility Organizations (PROs) in each EU destination country and report quantities sold. (ecosistant)

EPR Implementation Stage	Deadline	Requirement	Indian Exporter Action
National law transposition	June 2027	Member States transpose EPR into national law	Monitor national implementations
EPR schemes operational	April 2028	PROs operational, fees start applying	Register with PROs in each target market
Micro-enterprise extension	April 2029	EPR obligations apply to micro-enterprises	Prepare early if below current threshold
Eco-modulation active	2028+	Fees adjusted for durability, recyclability, recycled content	Design products for lower EPR fees

2.5 ECGT Directive: The Marketing Compliance Layer

The Empowering Consumers for the Green Transition (ECGT) Directive 2024/825, adopted in March 2024, amends the Unfair Commercial Practices Directive to directly address **greenwashing** in consumer-facing marketing. (greenstitch.io) EU member states must transpose ECGT into national law by **March 2026**, with enforcement beginning **September 2026**. The directive introduces three major prohibitions that directly affect how Indian textile exporters (and their EU brand customers) communicate sustainability claims.

First, **generic environmental claims are banned** unless the company can prove "recognised excellent environmental performance" such as achieving the EU Ecolabel. Terms like "eco-friendly," "green," "sustainable," "nature's friend," "conscious," and "responsible" are prohibited without specific, verifiable substantiation. (greenstitch.io) A t-shirt cannot be labelled "The Eco Tee" simply because it uses slightly less water — the claim must specify exactly what environmental benefit is being claimed and how it is measured. Second, **climate neutrality claims based on carbon offsetting are banned**. A brand cannot claim its polyester garments are "carbon neutral" because it purchased tree-planting credits. (greenstitch.io) Third, **self-certified sustainability labels are prohibited**. Brands cannot create their own "Green Leaf" icon — labels must be based on officially recognized certification schemes (GOTS, OEKO-TEX, EU Ecolabel) or established by public authorities.

For Indian manufacturers, ECGT compliance cascades through the supply chain. EU brand customers will increasingly demand that sustainability claims made about products are backed by **certifiable, testable data** that flows from the manufacturer. The platform's **ECGT Claim Validator** module addresses this by cross-referencing any marketing claim against the underlying certification data, test reports, and supply chain evidence, flagging claims that lack sufficient substantiation or that use prohibited generic language.

2.6 Product Architecture: Textile EU Compliance Platform

The Textile EU Compliance Platform mirrors the four-layer architecture of the Battery Passport Platform but is optimized for the textile-specific regulatory stack, data sources, and manufacturing workflows

prevalent in India's textile clusters (Tiruppur, Surat, Ludhiana, Panipat, Ichalkaranji).

2.6.1 Layer 1: Data Collection & Supplier Onboarding The textile data collection layer addresses the fundamental challenge of the Indian textile supply chain: extreme fragmentation. A single garment may pass through **15-20 different processing stages** across multiple facilities — spinning, knitting/weaving, dyeing, printing, finishing, cutting, sewing, washing, and packaging — each potentially operated by a different SME with minimal digital infrastructure. The platform's **Supplier Onboarding module** provides a guided, multi-step process for enrolling each supply chain partner, capturing facility information, process capabilities, certification status, and contact details. The onboarding workflow supports **Hindi, Tamil, Telugu, Gujarati, Punjabi, Marathi, Bengali, and English** to accommodate India's major textile-producing regions.

The **Chemical Inventory module** maintains a comprehensive database of all chemicals used at each processing stage, cross-referenced against the REACH restricted substance list, SVHC candidate list, and OEKO-TEX restricted substance catalogue. For each chemical, the module stores SDS (Safety Data Sheet), supplier declaration, concentration limits, and test reports. The module automatically flags chemicals that appear on updated REACH restriction lists and alerts affected suppliers and manufacturers. It also supports **ZDHC (Zero Discharge of Hazardous Chemicals) MRSL** conformance tracking, which is increasingly required by major EU brands.

Material Composition data is captured at the fiber, yarn, fabric, and finished product levels. The platform supports BOM (Bill of Materials) import from ERP systems, fiber content testing data upload (from certified labs like SGS, Intertek, TUV), and recycled content verification through GRS (Global Recycled Standard) or RCS (Recycled Claim Standard) certification chain-of-custody documentation. The material composition engine calculates weighted averages across blended products and generates the standardized fiber content declarations required for EU labelling.

Data Source	Integration Method	Data Captured	Region-Specific Consideration
Supplier Onboarding Portal	Web app + mobile	Facility info, certifications, process capabilities	Multi-language; low-bandwidth optimization for rural areas
Chemical Inventory	SDS upload + chemical database	Substance CAS numbers, concentrations, hazard classes	ZDHC MRSL alignment; REACH auto-flagging
Material Composition	ERP import + lab test upload	Fiber content, recycled %, origin data	GRS/RCS chain-of-custody tracking
Certification Tracker	API + manual upload	GOTS, OEKO-TEX, BCI, Fair Trade, ISO 9001	Auto-renewal alerts; scope verification
Production Data	ERP connector (Tally, SAP, etc.)	Order volumes, SKUs, destinations, unsold inventory	Tiruppur/Surat cluster-specific ERP adaptations
Testing Reports	Lab portal API + PDF upload	REACH testing, azo dye testing, heavy metal testing	Integration with SGS, Intertek, Bureau Veritas India

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2.6.2 Layer 2: Compliance Engine The Textile Compliance Engine runs five specialized modules, each addressing a distinct regulatory requirement in the textile stack.

The **REACH Analyzer** continuously monitors the chemical inventory against the latest REACH restrictions, SVHC candidate list updates, and national implementations (Germany's BfR, France's DGCCRF). When a substance is newly restricted or a threshold changes, the analyzer identifies all affected products, batches, and suppliers, generates risk assessments, and triggers remediation workflows. The analyzer also produces **SCIP database notifications** (Substances of Concern In articles as such or in complex objects) required under the Waste Framework Directive for SVHCs above 0.1% w/w.

The **ESPR Ecodesign Checker** evaluates products against the expected ecodesign requirements from the JRC preparatory study. While the delegated act is not yet finalized, the checker implements the proposed requirements as configurable rules: recycled content thresholds (20% cotton, 15% polyester, 10% wool, 5% nylon), durability scoring based on construction and fiber properties, repairability assessment (availability of spare parts, construction enabling repair), and manufacturing footprint benchmarking. (carbonfact.com) Products that meet proposed thresholds receive a "Likely Compliant" pre-assessment, giving manufacturers early visibility into potential gaps.

The **EPR Fee Calculator** estimates textile EPR fees for each EU member state where the manufacturer sells. Because EPR schemes are implemented nationally, fee structures vary by country. The calculator incorporates known fee models (France's Refashion, Netherlands' NV Afvalfonds) and projected models for countries still developing schemes (Spain, Italy, Germany). It applies eco-modulation factors — products with higher recycled content, better durability scores, and improved recyclability receive fee discounts. The calculator enables manufacturers to **optimize product design for lower EPR costs**, turning compliance into a cost-saving exercise.

The **Destruction Ban Monitor** tracks unsold inventory positions across EU distribution channels, flags inventory at risk of destruction, and generates compliance documentation for permitted destruction scenarios (derogations). The module integrates with warehouse management systems and EU retailer inventory feeds to provide real-time visibility into stock positions. When inventory cannot be sold through regular channels, the module generates **donation workflows** — automated outreach to social economy entities, website listing for surplus goods, and documentation of all attempts required for derogation compliance.

The **ECGT Claim Validator** audits all consumer-facing marketing claims against the underlying evidence base. The module parses product descriptions, hangtag text, website content, and collection names, identifies environmental and sustainability claims, and cross-references each claim against available certifications, test data, and supply chain evidence. Claims using prohibited generic terms are flagged. Claims based on offsetting are flagged. Self-created sustainability labels are flagged. The module generates **claim substantiation reports** that EU brand customers can use to defend their marketing against regulatory challenge.

2.6.3 Layer 3: DPP Generator & Sustainability Scoring The DPP Generator creates the textile Digital Product Passport in anticipation of the delegated act requirements. Based on the JRC milestone 3 proposals and ESPR framework, the generator produces passports containing product identification, material composition, manufacturing footprint, circularity attributes, chemical compliance status, supply chain transparency data, and EPR categorization. (carbonfact.com) The passport is accessible via **GS1 Digital Link QR code** printed on the care label or hangtag, enabling consumer scanning for sustainability information and regulatory scanning for compliance verification.

The **Sustainability Scoring engine** goes beyond compliance to create competitive differentiation. It calculates a **Composite Sustainability Index** for each product based on: carbon intensity (kg CO2e per garment), water consumption (liters per garment), chemical hazard score (based on REACH restrictions avoided), circularity score (recycled content + recyclability + durability), and social compliance score (based on supplier audit results). Products scoring in the top quartile receive a visual "EU Ready" badge that manufacturers can use in B2B communications with EU buyers. This scoring system anticipates the eco-modulation mechanisms in EPR and the performance class systems likely to emerge under ESPR.

The **Circularity Index** specifically addresses the circular economy requirements embedded across the textile regulatory stack. It measures: (1) **Design for longevity** — fiber quality, construction durability, color fastness; (2) **Design for recycling** — mono-material construction, detachable trims, fiber identification; (3) **Recycled input** — pre-consumer and post-consumer recycled content percentage; (4) **End-of-life pathways** — recyclability assessment, biodegradability (where applicable), and take-back program participation. Products with high circularity scores will benefit from lower EPR fees and may qualify for preferential treatment under future EU procurement rules.

2.6.4 Layer 4: Registry Integration & Multi-Country Reporting The registry integration layer for textiles is more complex than for batteries because textile compliance is fragmented across **27 national EPR schemes**, each with different registration requirements, fee structures, and reporting formats. The platform maintains a **Country Compliance Matrix** that tracks the status of each EU member state's textile EPR implementation and automatically configures reporting workflows when new national schemes go live.

The **EU DPP Registry Connector** handles submission of textile product identifiers and passport URLs to the central EU DPP Registry (live July 2026). The **National EPR Portal Connectors** integrate with each country's Producer Responsibility Organization (PRO) registration and reporting systems — France's Refashion, Netherlands' NV Afvalfonds, and future systems in Germany, Spain, Italy, and others. The **ECGT Compliance Reporter** generates documentation demonstrating that marketing claims are substantiated and that no prohibited generic terms are used. The **Unsold Goods Reporter** formats and submits the standardized disclosure reports on discarded unsold products, required from February 2027 for large companies. (anthesisgroup.com)

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3. Market Opportunity & Competitive Landscape

3.1 Addressable Market

The combined addressable market for compliance services targeting Indian battery and textile exporters to the EU is substantial and growing rapidly. India's textile and apparel exports to the EU reached approximately **$7.6 billion in FY2024-25**, with the EU representing roughly **19% of India's total textile export basket**. (Apparel Views) The India-EU Free Trade Agreement, signed in July 2025 and operational in early 2026, eliminates the **9-12% import duties** that previously disadvantaged Indian textiles against competitors like Bangladesh and Vietnam. (Keypoint Intelligence) Industry projections suggest Indian textile exports to the EU could reach **$15 billion by 2030** under the FTA's zero-duty regime. (The Economic Times) India's lithium-ion battery export market, while currently smaller at approximately **$320-480 million annually**, is growing at **40%+ CAGR** driven by the PLI scheme and EU demand for energy storage and EV batteries. (Volza.com)

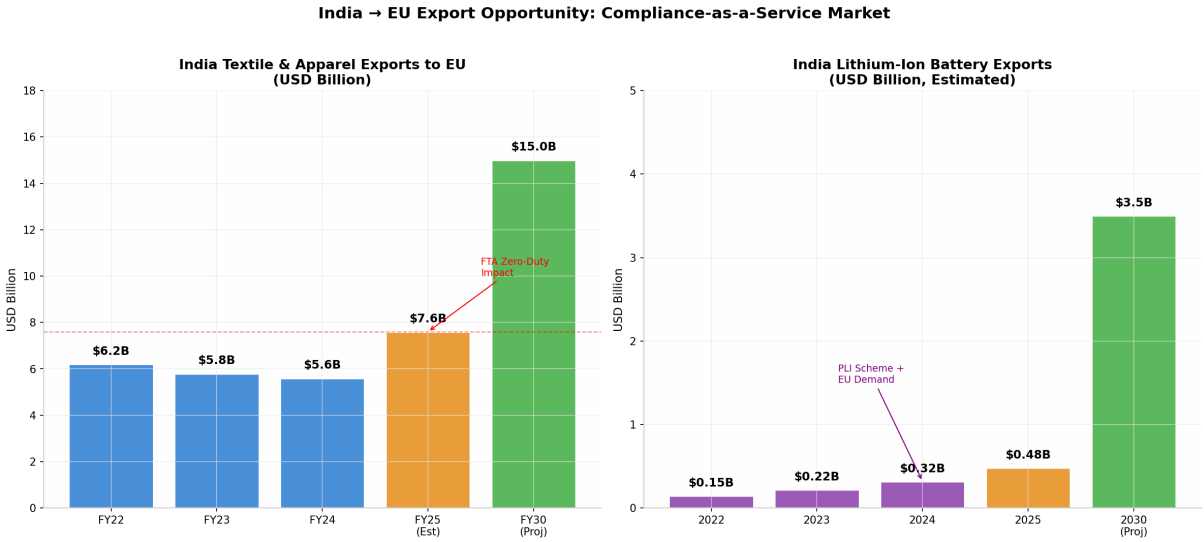


Figure 4: Market Opportunity

The **Digital Product Passport platform market globally** is projected to reach **$2.4 billion** by 2025, with the competitive landscape including specialized players like **Circularise** (13% market share, blockchain-based traceability), **Everledger**, **Minespider**, and enterprise incumbents like **SAP, Siemens, IBM,** and **TCS/Infosys** from India. (Future Market Insights) However, **no existing platform is purpose-built for Indian SME manufacturers** navigating EU compliance. Current solutions are either (a) designed for large EU-based enterprises with existing IT infrastructure, (b) focused on a single regulation rather than the integrated stack, or (c) priced beyond the reach of India's SME-dominated export base. The product opportunity lies in building a **India-first, EU-compliant** platform that combines regulatory depth with affordability, local language support, and integration with India's domestic compliance systems (CPCB EPR, BIS, RoSCTL).

Market Segment	Current Size	Growth Rate	Key Driver	Compliance Spend Potential
Indian textile exports to EU	$7.6B (FY25)	15-20% CAGR	India-EU FTA zero-duty	$50-150M annually (0.5-2% of export value)
Indian battery exports to EU	$0.48B (2025)	40%+ CAGR	EV + energy storage demand	$10-30M annually (2-6% of export value)
DPP platform market (global)	$2.4B (2025)	35% CAGR	ESPR + Battery Regulation	N/A
Indian SME compliance services	$200M (est.)	25% CAGR	Regulatory complexity increase	Platform capture: $20-50M by 2030

3.2 Competitive Differentiation

The proposed dual-product platform differentiates from existing solutions across five dimensions that are specifically tailored to the Indian manufacturing context.

Differentiation Dimension	Existing Solutions (Global)	Our Platform (India-First)
Regulatory Scope	Single regulation focus (REACH-only, or DPP-only)	Integrated stack: ESPR + REACH + EPR + ECGT + CSRD + Indian regulations
Target User	Large enterprises with dedicated compliance teams	SME-first with enterprise tier; designed for limited compliance staff
Language & Localization	English-only; EU-centric terminology	8 Indian languages; India-EU regulatory bridge terminology
Domestic Integration	EU-only systems	CPCB EPR, BIS, PLI, RoSCTL integration alongside EU compliance
Pricing Model	Enterprise SaaS ($50K-$500K/ year)	Tiered: Self-service ($200/month) → Professional ($2K/month) → Enterprise (custom)

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4. Shared Platform Infrastructure

While the Battery Passport and Textile EU Compliance products serve different regulatory domains, they share a common technical infrastructure that enables economies of scale, cross-product insights, and a unified user experience for manufacturers operating in both categories.

4.1 Common Technical Stack

Infrastructure Layer	Technology Choice	Rationale
Cloud Platform	AWS Mumbai / Azure India Central	Data residency compliance; low latency for Indian users; DR in EU (Frankfurt/Ireland)
Database	PostgreSQL (structured) + MongoDB (document) + Neo4j (graph)	Relational for compliance records; document for flexible DPP schemas; graph for supply chain
API Gateway	Kong / AWS API Gateway	Rate limiting, authentication, request routing for all external integrations
Identity & Access	OAuth 2.0 + OIDC + RBAC	Multi-tenant isolation; role-based permissions for public/restricted DPP data
Blockchain Anchoring	Ethereum L2 (Polygon) / EBSI	Tamper-evident audit trails for critical compliance records; EU EBSI alignment
QR Code / GS1	GS1 Digital Link resolver	Single QR code serving multiple audiences; dynamic content resolution

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Table 14 – continued

Infrastructure Layer	Technology Choice	Rationale
Analytics	Apache Superset + Python (Pandas/Scikit)	Compliance dashboards, predictive analytics, carbon optimization
Mobile	React Native	Supplier portal app for low-bandwidth Indian manufacturing environments

4.2 Cross-Product Synergies

Manufacturers that use both products benefit from **shared supplier profiles** — a chemical supplier serving both battery and textile divisions only needs to onboard once. **Carbon accounting methodologies** are harmonized across both products, enabling consolidated corporate carbon reporting. The **EU DPP Registry connector** is shared infrastructure, reducing per-product integration costs. **Training and support** resources (helpdesk, documentation, webinars) serve both user bases with regulatory updates that often intersect (ESPR developments affect both product categories).

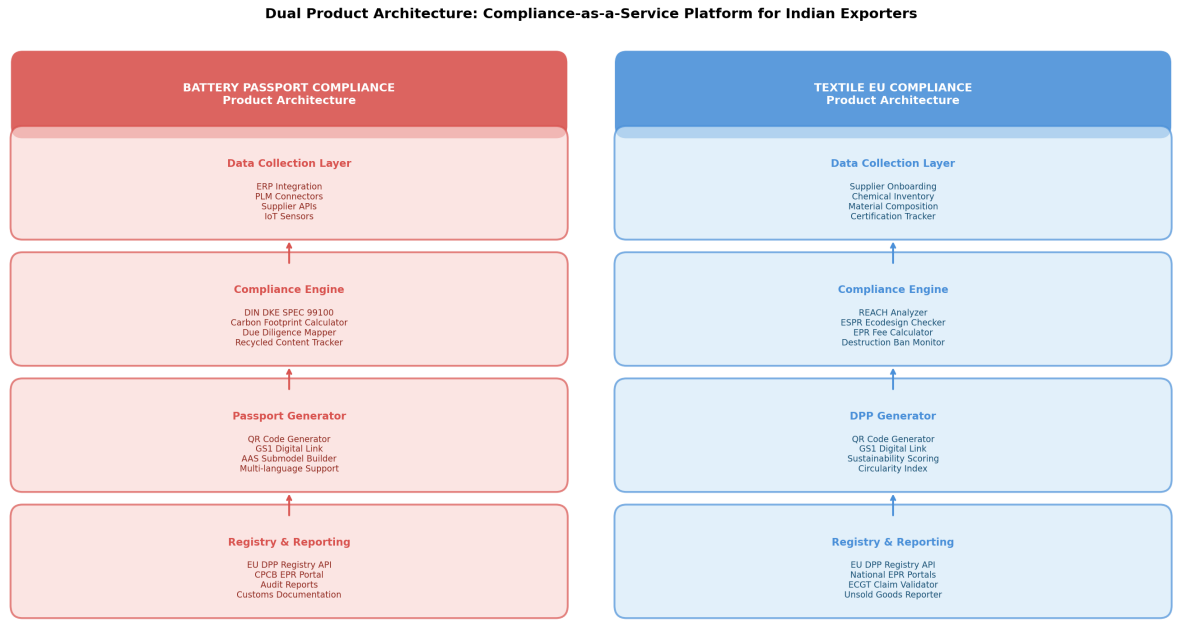


Figure 5: Product Architecture

5. Risk Assessment & Mitigation

5.1 Regulatory Risk

Risk	Likelihood	Impact	Mitigation
Delegated act delays push back DPP deadlines	Medium (pattern observed)	Reduces urgency, extends runway	Build flexible data models; configurable rule engine
Omnibus II further simplifies CSRD/CSDDD	Medium	Reduces reporting burden for some customers	Monitor legislative process; adjust ESRS module scope
India-EU FTA implementation delays	Low	Slower export growth, smaller addressable market	Diversify to UK, Middle East, ASEAN compliance modules
EU customs enforcement inconsistent across member states	High	Uncertain compliance ROI perception	Focus on Germany, Netherlands, France (strictest enforcement)

5.2 Market Risk

Risk	Likelihood	Impact	Mitigation
Large Indian manufacturers build in-house solutions	Medium	Lost enterprise revenue	Focus on SME segment; provide superior time-to-value
Global players (SAP, Siemens) enter Indian market	High	Price competition, brand disadvantage	Differentiate on India-specific features, language, domestic integration
Low digital maturity among Indian SME suppliers	High	Slow adoption, high support costs	Offline-capable tools, Excel import, mobile-first design, assisted onboarding

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6. Implementation Roadmap & Success Metrics

6.1 Development Phases

Quarter	Battery Product Milestone	Textile Product Milestone	Shared Infrastructure
Q3 2026	MVP: Supplier portal, basic passport, QR codes	MVP: REACH analyzer, supplier onboarding, chemical inventory	Cloud setup, identity system, GS1 resolver
Q4 2026	Carbon footprint calculator, DIN DKE SPEC mapper	EPR fee calculator, DPP generator (draft spec)	EU DPP Registry connector, blockchain anchoring
Q1 2027	AAS submodel export, due diligence mapper	ESPR ecodesign checker, ECGT validator	Analytics dashboard, mobile app v1

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Table 17 – continued

Quarter	Battery Product Milestone	Textile Product Milestone	Shared Infrastructure
Q2 2027	EU DPP Registry integration, audit reporting	National EPR portal integrations, destruction ban monitor	Multi-language expansion, AI-assisted gap analysis
Q3 2027	IoT/BMS integration, dynamic data	Sustainability scoring, circularity index	Cross-product insights, consolidated reporting

6.2 Key Performance Indicators

KPI	Year 1 Target	Year 3 Target	Measurement
Paying customers (both products)	50	500	Active subscription count
Passports/DPPs generated	10,000	500,000	Cumulative count
Supplier portal users	500	10,000	Registered supplier accounts
Compliance score average	60%	90%	Average passport/DPP completeness
Customer NPS	30	50	Quarterly survey
Revenue (ARR)	$500K	$8M	Monthly recurring revenue x 12

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7. Conclusion: The Compliance Imperative

The EU's regulatory transformation is not a temporary disruption — it is a **permanent structural shift** in how products enter the world's second-largest market. Indian manufacturers that treat compliance as a cost center will face margin compression, market access denial, and competitive displacement. Those that treat compliance as a strategic capability — enabled by purpose-built digital infrastructure — will capture market share from less-prepared competitors, command premium pricing from sustainability-conscious EU buyers, and build defensible competitive moats rooted in transparent, verifiable supply chain data.

The Battery Passport Compliance Platform and Textile EU Compliance Platform presented in this report are designed to transform regulatory burden into competitive advantage. By automating the collection, validation, and reporting of compliance data across the full EU regulatory stack, these products enable Indian manufacturers to focus on what they do best — making world-class products — while the platform handles the complexity of EU market access. The window for preparation is narrow: **February 2027 for batteries, late 2028 for textiles**. The cost of being early is minimal. The cost of being late is market exclusion from a **$20+ billion** export opportunity.